JAEKWON IM

Ph.D. candidate @ Music and Audio Computing Lab, KAIST

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Homepage | Google Scholar

I am a Ph.D. candidate at KAIST (Music and Audio Computing Lab) advised by Professor Juhan Nam. My research focuses on improving the quality, efficiency, and controllability of generative models for audio generation and processing. I am passionate about developing audio generative models that achieve high perceptual quality in real-world scenarios, and I have worked on tasks such as video-to-audio generation, audio super-resolution, and acoustic transfer. Previously, I co-founded AudAI, where I contributed to the development of singing voice synthesis, voice conversion, and neural vocoder.

RESEARCH INTERESTS

Audio Generation, Multimodal Learning, Audio Enhancement, Music Information Retrieval

EXPERIENCE

NVIDIA, Taipei, Taiwan — Mar 2026 - present

Research Intern

- Mentor: Sung-Feng Huang, Szu-Wei Fu
- Research on LLM-based speech generation

Sony CSL, Tokyo, Japan (Remote) — Jul 2025 - Sep 2025

Research Collaborator

- Research on video-to-audio/music generation, leading to *PF-D2M* (INTERSPEECH 2026)

AudAi, Seoul, South Korea — May 2023 - Jul 2025

Co-founder & AI / SW Engineer

- Contributed to the research and development of VOX Factory, an online singing voice synthesizer [demo]
- AudAi provided AI voice technology for SM Entertainment's virtual artist nævis [demo]
- AI: neural vocoder, singing voice synthesis/conversion — Frontend: SolidJS

DCASE 2023 — Dec 2022 - Jun 2023

Organizer of Challenge task 7 (Foley Sound Synthesis)

- Prepared baseline model and training dataset
- Developed submission template, and co-managed the challenge pipeline

Gaudio lab, Seoul, South Korea (Remote) — Nov 2022 - Feb 2023

Internship in AI group

- Proposed and co-organized the DCASE 2023 Foley Sound Synthesis Challenge

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — 2022 - present

Ph.D. candidate in Graduate School of Culture Technology

Music and Audio Computing Lab (Advisor: Juhan Nam)

Korea Advanced Institute of Science and Technology (KAIST) — 2020 - 2022

M.S. in Graduate School of Culture Technology

Music and Audio Computing Lab (Advisor: Juhan Nam)

Chungnam National University — 2013 - 2020

B.E. in Computer Science and Engineering

PUBLICATIONS

PF-D2M: A Pose-free Diffusion Model for Universal Dance-to-Music Generation
Jaekwon Im, Natalia Polouliakh, Taketo Akama
Proceedings of the 27th INTERSPEECH Conference (Interspeech), 2026

SAGA-SR: Semantically and Acoustically Guided Audio Super-Resolution
Jaekwon Im and Juhan Nam
Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026

Video-Foley: Two-Stage Video-To-Sound Generation via Temporal Event Condition for Foley Sound
Junwon Lee, **Jaekwon Im**, Dabin Kim, Juhan Nam
IEEE/ACM Transactions on Audio, Speech and Language Processing (TASLP), 2025

FlashSR: One-step Versatile Audio Super-resolution via Diffusion Distillation
Jaekwon Im and Juhan Nam
Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025

DIFFRENT: A Diffusion Model for Recording Environment Transfer of Speech
Jaekwon Im and Juhan Nam
Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024

Foley Sound Synthesis at the DCASE 2023 Challenge
Keunwoo Choi∗, **Jaekwon Im**∗, Laurie M. Heller∗, Brian McFee, Keisuke Imoto, Yuki Okamoto, Mathieu Lagrange, Shinnosuke Takamichi (∗ equal contribution)
Proceedings of the 8th Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE), 2023

Neural Vocoder Feature Estimation for Dry Singing Voice Separation
Jaekwon Im, Soonbeom Choi, Sangeon Yong, Juhan Nam
Proceedings of the 14th Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA), 2022

ADVISING, TEACHING & SERVICE

- Reviewer, TASLP, ICASSP
- TA, GCT634 Musical Applications of Machine Learning (Mar 2024 - Jun 2024)
- TA, CTP431 Fundamentals of Computer Music (Sep 2023 - Dec 2023)
- TA, GCT731 Topics in Music Technology<Generative AI for Music> (Mar 2023 - Jun 2023)
- Mentor, Daejeon Science High School Research & Education (Mar 2022 - Dec 2022)
- TA, GCT634 Musical Applications of Machine Learning, KAIST (Sep 2022 - Dec 2022)
- TA, GCT535 Sound Technology for Multimedia, KAIST (Mar 2022 - Jun 2022)

AWARDS

- KAIST Lab Startup 2nd Place Excellence Prize (2022)
- E*5 KAIST Final Development Award (2021)
- Korea Computer Congress (KCC) 2019 encouragement award (2019)
- CNU 'Thinking Programming Competition' first prize (2019)

LANGUAGES & SKILL & INTERESTS

- English(fluent), Korean(native)
- Python (Pytorch), Solid.js, React.js, HTML/CSS, C++, Java
- Music producing